

Customer Shopping Behavior Analysis

Uncovering insights from **3,900 transactions** to guide strategic business decisions.



Dataset Overview

3,900

Total Purchases

Comprehensive transaction data

18

Data Columns

Customer demographics & behavior

37

Missing Values

Only in Review Rating column

25

Unique Items



Data Preparation Process

Clean & Standardize

Imputed missing ratings by category median; renamed columns to snake_case

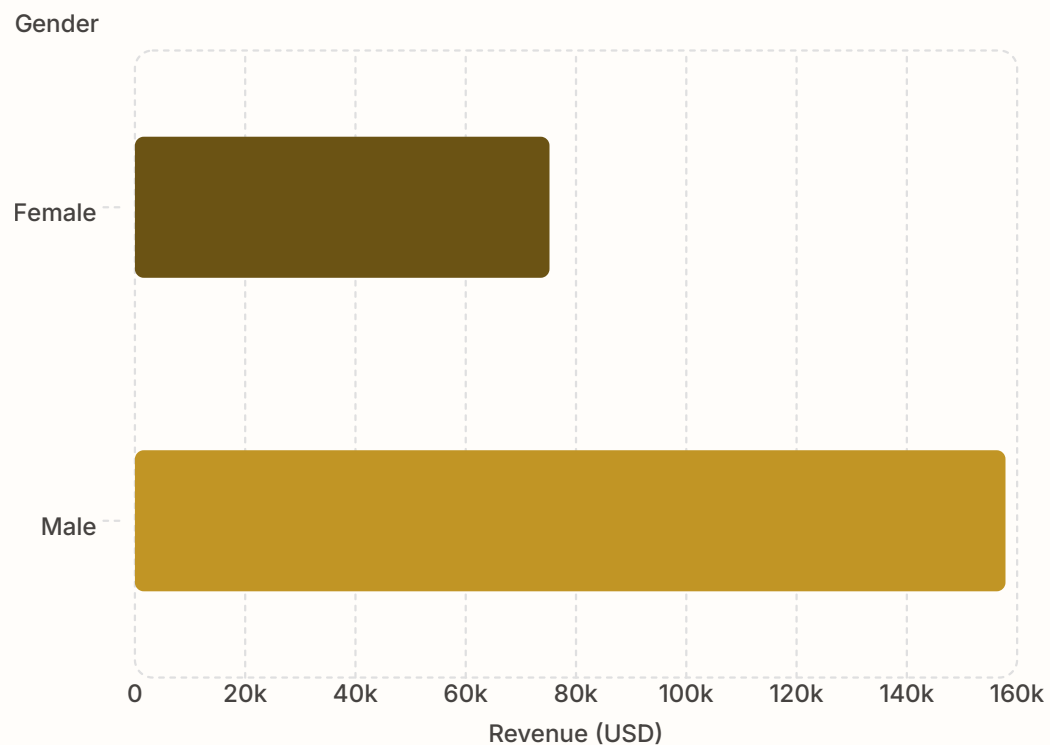
Feature Engineering

Created age_group bins and purchase_frequency_days

Database Integration

Loaded cleaned DataFrame into PostgreSQL for SQL analysis

Revenue by Gender



Key Insight

Male customers generated **2.1× more revenue** than female customers — a significant gap worth investigating.

i Male: \$157,890 | Female: \$75,191

Top Products & Discount Patterns

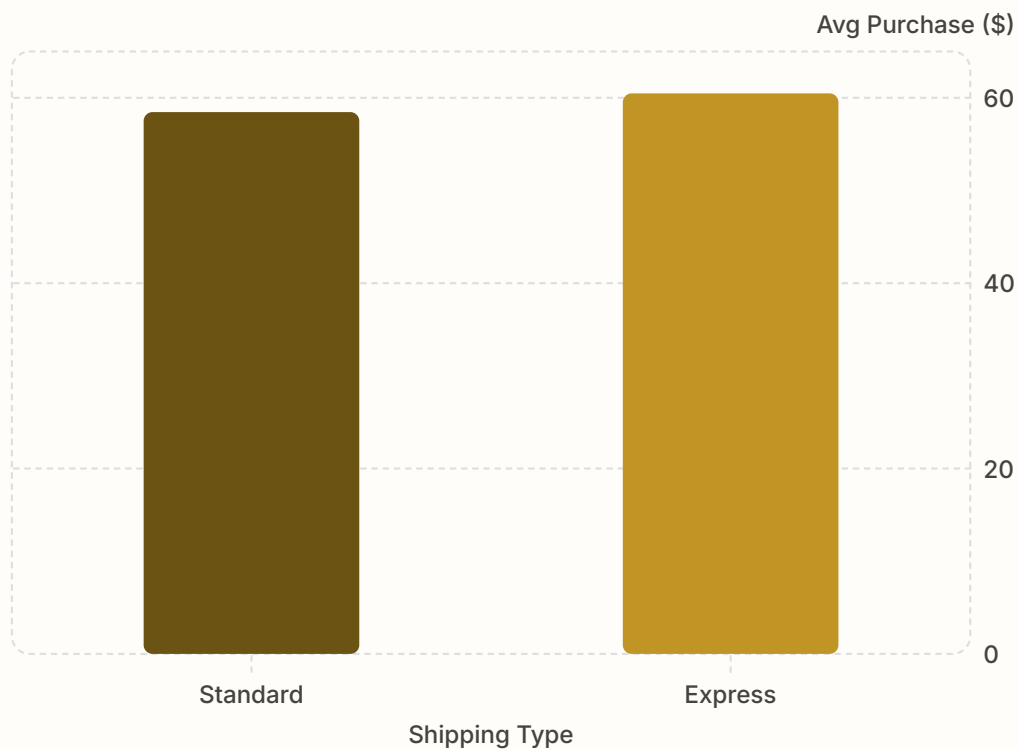
Highest-Rated Products

Gloves	3.86 ★
Sandals	3.84 ★
Boots	3.82 ★
Hat	3.80 ★
Skirt	3.78 ★

Most Discount-Dependent

Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

Shipping & Subscription Insights



Subscribers vs. Non-Subscribers

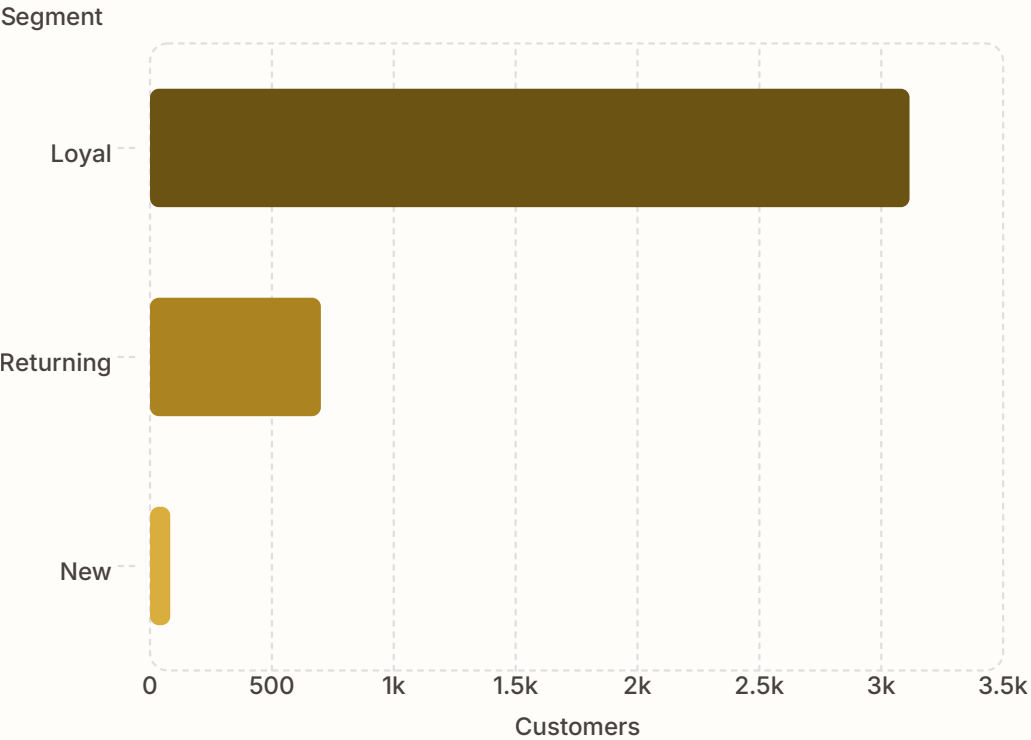
Subscribers (27%)

1,053 customers · \$59.49
avg · \$62,645 total

Non-Subscribers (73%)

2,847 customers · \$59.87
avg · \$170,436 total

Customer Segmentation



Segment Breakdown

Loyal — 80%

3,116 customers

Returning — 18%

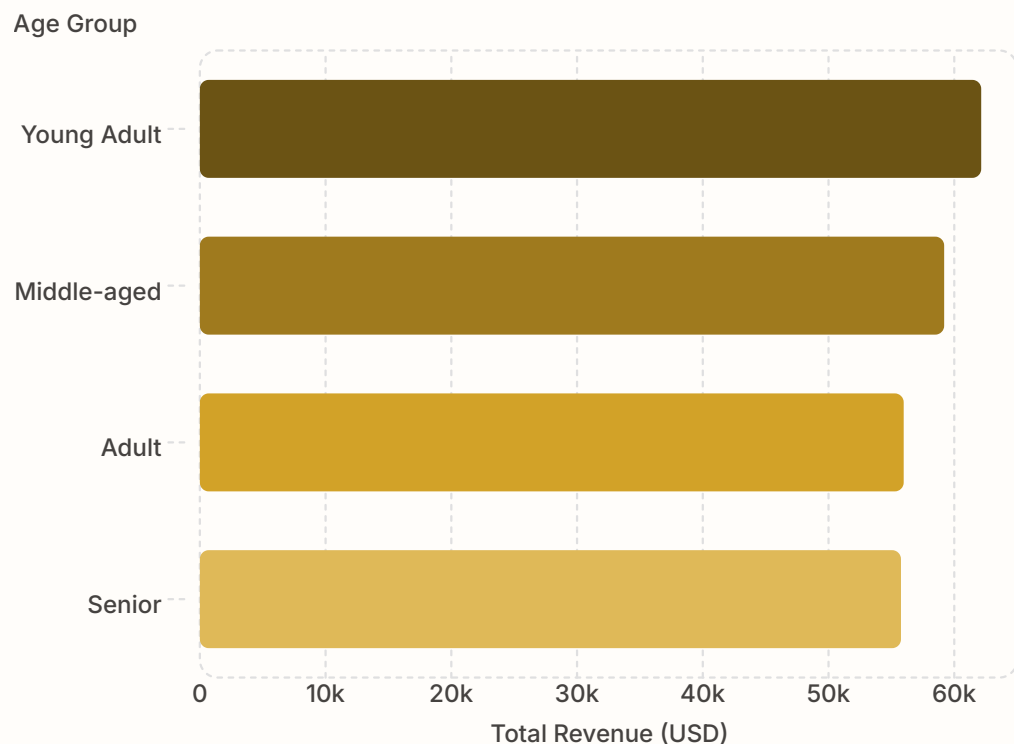
701 customers

New — 2%

83 customers

Of 958 repeat buyers (5+ purchases), most are not subscribers.

Revenue by Age Group



Age Group Performance

Young Adults lead in revenue contribution. Revenue is relatively balanced across groups — suggesting broad market appeal.

📌 **Young Adults** generate the highest total revenue at \$62,143.

Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits



Loyalty Programs

Reward repeat buyers toward Loyal segment



Review Discount Policy

Balance sales boosts with margin control



Targeted Marketing

Focus on high-revenue age groups & express-shipping users

